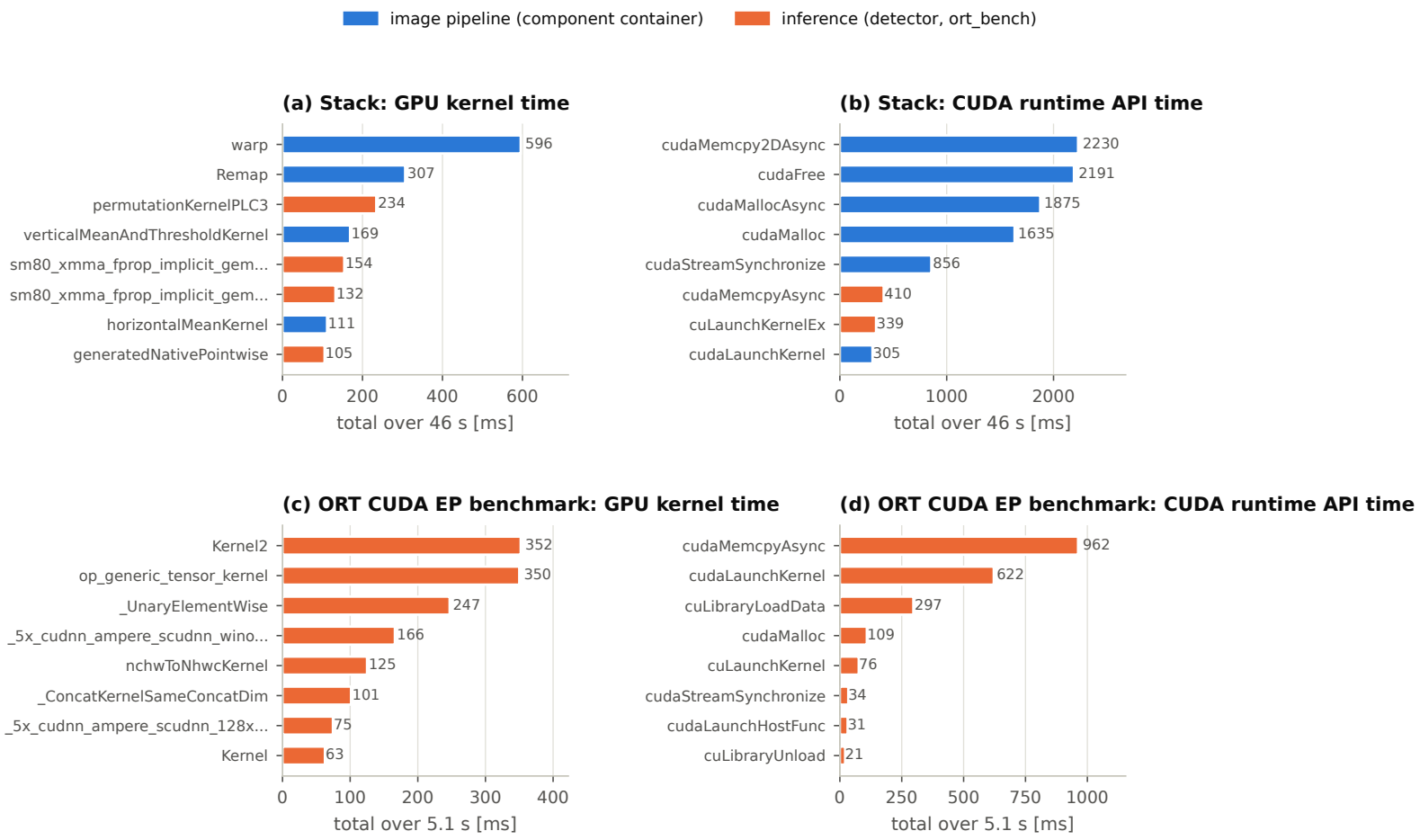
image pipeline (component container)
inference (detector, ort_bench)
(a) Stack: GPU kernel time
warp
Remap
permutationKernelPLC3
verticalMeanAndThresholdKernel
sm80_xmma_fprop_implicit_gem...
sm80_xmma_fprop_implicit_gem...
horizontalMeanKernel
generatedNativePointwise
596
307
234
169
154
132
111
105
0
200
400
600
total over 46 s [ms]
(b) Stack: CUDA runtime API time
cudaMemcpy2DAsync
cudaFree
cudaMallocAsync
cudaMalloc
cudaStreamSynchronize
cudaMemcpyAsync
cuLaunchKernelEx
cudaLaunchKernel
2230
2191
1875
1635
856
410
339
305
0
1000
2000
total over 46 s [ms]
(c) ORT CUDA EP benchmark: GPU kernel time
Kernel2
op_generic_tensor_kernel
_UnaryElementWise
_5x_cudnn_ampere_scudnn_wino...
nchwToNhwcKernel
_ConcatKernelSameConcatDim
_5x_cudnn_ampere_scudnn_128x...
Kernel
352
350
247
166
125
101
75
63
0
100
200
300
400
total over 5.1 s [ms]
(d) ORT CUDA EP benchmark: CUDA runtime API time
cudaMemcpyAsync
cudaLaunchKernel
cuLibraryLoadData
cudaMalloc
cuLaunchKernel
cudaStreamSynchronize
cudaLaunchHostFunc
cuLibraryUnload
962
622
297
109
76
34
31
21
0
250
500
750
1000
total over 5.1 s [ms]